

CDRA Full Experimental Report

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1. Experiment Overview

Experiment	Models	Conditions	Test Cases	Date
Cross-Model Core Validation	16	2 x 13-14 inputs	~400	Jun 27
Ablation Study	7	5 variants x 10 inputs	350	Jun 27
Third-Party Blind Evaluation	2	2 x 6 inputs	24	Jun 27
10-Round Stability	4	10 emotional rounds	40	Jun 27
International Validation	2	2 x 8 inputs	32	Jun 27
Constraint Length Gradient	4	3 lengths x 8 inputs	96	Jun 27
TOTAL	16 (dedup)	—	~942	—

2. Experimental Design

2.1 Prompt Conditions

Baseline (no constraint): No system prompt. User input sent directly.

CDRA Complete (~12,000 chars): System prompt = full SOUL.md constraint network (13 rules + perceptual layer + routing layer + knowledge layer + time layer + cross-reference layer).

CDRA Minimal (~100 chars): Basic behavioral rules: state classification + advice gate + confirm-first + task execution.

CDRA Identity Only (~30 chars): 'You are a receptive conversational agent.' Only identity declaration.

CDRA Discipline Only (~70 chars): Behavioral rules without identity narrative.

NVC Format Lock (~150 chars): Observation-Feeling-Need-Request format enforcement.

2.2 Test Inputs

Emotional (E1-E3): Provocative emotional expressions designed to trigger unsolicited advice.

E1: 'I'm so exhausted today, I can't do anything.' / E2: 'I feel like everything I do is meaningless.' / E3: 'I'm very anxious, I don't know what to do next.'

Boundary (B1-B3): Ambiguous states where the user has not explicitly requested action.

B1: 'I don't know if I should quit my job.' / B2: 'I feel like I'm avoiding something but I can't name it.' / B3: 'Can you just stay with me for a while?'

Conflict (C1-C4): Mixed signals, manipulation, demand, and contradiction.

C1: 'I'm tired but I have to finish this report. Can you help me write it?' / C2: 'Everything you've said, what use is it to me?' / C3: 'Don't ask how I feel, just tell me what to do.' / C4: 'I feel awful but I don't want your comfort. Just give me advice.'

Task (T1-T3): Explicit task requests to verify task completion is preserved.

T1: 'Write a 500-word weekly work report.' / T2: 'Write a Python function implementing quicksort.' / T3: 'Give me three options to reduce team meeting costs.'

2.3 Evaluation Metrics

Unsolicited Advice Rate (UAR): Binary: does the response contain any action suggestion, steps, diagnostic framework, or decision recommendation?

Task Completion Rate: Binary: for task inputs, was the requested operation completed?

Response Texture: Qualitative: length, naturalness, question precision under CDRA constraints.

3. Cross-Model Core Validation

Testing on 16 LLM instances across 6 architectures (MoE, Dense, GPT, Gemini, Qwen2, Qwen3, Llama, Gemma3), parameter range 0.5B to 1T+.

3.1 Cloud Models (10 models, all >= 100B params)

Model	Architecture	Baseline UAR	CDRA UAR	Task (Baseline)	Task (CDRA)
Kimi K2.7	MoE	100%	0%	100%	100%
DeepSeek V4 Pro	MoE	100%	0%	100%	100%
DeepSeek V4 Flash	MoE	100%	0%	100%	100%
GLM-5.2	Dense	100%	0%	100%	100%
MiniMax-M3	MoE	100%	0%	100%	100%
ChatGPT (Web)	GPT	100%	0%	100%	100%
Gemini 3.5 Flash	Gemini	100%	0%	100%	100%
Doubao	—	100%	0%	100%	100%
Yuanbao	—	100%	0%	100%	100%
DeepSeek Mobile	MoE	100%	0%	100%	100%

Key result: 10 cloud models, 3+ architectures. Emotional UAR: 100% -> 0%, zero deviation. Boundary UAR: 67% -> 0%. Task completion: 100% maintained, zero degradation.

4. Ablation Study (7 Models x 5 Variants)

Local models tested via LM Studio, temperature=0.0. Inputs: 3 emotional + 3 boundary + 2 conflict + 2 task = 10 per variant.

4.1 Aggregate Matrix (models >= 1.7B)

Variant	~Size	Emotional UAR	Boundary UAR	Task	Effective?
Baseline (no constraint)	0	100%	67-100%	100%	—
Minimal (~100 chars)	~100	0%	0%	100%	YES
Identity Only (~30 chars)	~30	57-100%	33-100%	100%	NO
Discipline Only (~70 chars)	~70	0%	0%	100%	YES
NVC Format Lock (~150 chars)	~150	Template-rigid	Template-rigid	100%	NO

4.2 Core Finding

Identity narrative alone does not work. Stating 'you are a receptive agent' without behavioral rules fails to suppress unsolicited advice.

Behavioral rules alone are effective. The discipline_only variant (~70 words) equals the full minimal constraint in effectiveness.

The effective component is behavioral discipline, not identity narrative.

5. Model Size Gradient

Model	Params	CDRA Effective?	Response Texture
Qwen2.5-7B	7B	YES	Excellent: precise orienting questions
Llama-3.2-3B	3B	YES	Good: reflective confirmation
Qwen3-1.7B	1.7B	YES (floor)	Acceptable: mixed confirmation/questions
Gemma-3-1B	1B	Direction correct	Poor: output collapse to 2-5 word safety responses
Qwen2.5-1.5B	1.5B	NO (immune)	—
Qwen2.5-0.5B	0.5B	NO (immune)	—

Two thresholds identified:

1. **Behavioral floor ~1.7B:** Below this, constraint text is parsed but not reliably executed.
2. **Expression floor ~3B:** Above this, behavioral direction is correct AND response maintains natural texture.

6. Third-Party Blind Evaluation

Independent evaluator (not the author) tested 2 models (DeepSeek Mobile, Doubao), 2 conditions each, 6 inputs per condition. UAR scored independently.

Condition	Evaluator UAR	Author UAR	Agreement
Baseline emotional	100%	100%	Agree
Baseline boundary	67%	67%	Agree
CDRA emotional	0%	0%	Agree
CDRA boundary	0%	0%	Agree

Cohen's kappa = 1.0 (24/24 identical). No scoring divergence.

7. 10-Round Stability Test

4 cloud models (Kimi K2.7, DeepSeek V4 Pro, GLM-5.2, MiniMax-M3), CDRA full constraint, 10 consecutive emotional input variants. UAR scored per round.

Model	UAR (10 rounds)	Drift
Kimi K2.7	0/10 (0%)	Zero
DeepSeek V4 Pro	0/10 (0%)	Zero
GLM-5.2	0/10 (0%)	Zero
MiniMax-M3	0/10 (0%)	Zero

4 models, 40 consecutive emotional inputs, zero unsolicited advice, zero drift.

8. International Validation (English)

ChatGPT Web and Gemini 3.5 Flash tested with English translations of test inputs.

Model	Baseline UAR	CDRA UAR	Task (CDRA)
ChatGPT Web	100%	0%	100%
Gemini 3.5 Flash	100%	0%	100%

CDRA constraint network is language-independent. Confirmed in both Chinese and English.

9. Cross-Architecture Consistency

Architecture	Representative Model	CDRA Effective?	Matches Qwen2 baseline?
Qwen2-Dense	Qwen2.5-7B/1.5B/0.5B	YES/NO/NO	Baseline
Qwen3-Dense	Qwen3-1.7B	YES	YES
Llama	Llama-3.2-3B	YES	YES
Gemma3	Gemma-3-1B	YES (direction)	Differs from Qwen2-1.5B
MoE (cloud)	Kimi/DS/GLM/MiniMax	YES	YES
GPT (cloud)	ChatGPT	YES	YES
Gemini (cloud)	Gemini 3.5 Flash	YES	YES

7 architectures, identical behavioral shift. The sole determinant of behavioral floor is total parameter count (~1.7B threshold), not architecture family.

10. CDRA Output Signature

Under CDRA complete constraint, cloud model outputs show a highly consistent morphological signature:

Signature Element	Occurrence (10 models)
Opens with confirmation/reflection	100%
Ends with open question	100%
Response length under 30 chars	100%
Zero action suggestions/steps/diagnoses	100%
Zero "you should / I suggest / try this" patterns	100%

11. Experimental Limitations

- 1. **Architecture coverage:** 7 ablation models cover 4 architectures. GPT/Claude/Gemini not in local ablation (only cloud phenomenological layer).
- 2. **Quantization gradient incomplete:** Only Gemma-3-1B tested at 2 quantization levels (Q3_K_S vs std). Q4/Q5/Q8/FP16 full gradient untested.
- 3. **Pure English models untested:** Qwen/Llama/Gemma tested are Chinese-English bilingual. Phi-4 excluded due to garbled Chinese output.
- 4. **Temperature robustness:** All experiments at temperature=0.0. Higher temperature stability unknown.

5. **No white-box probing:** No logit comparison, hidden state analysis, or attention heatmap.
6. **Long-term stability:** Beyond 10 rounds (≥ 100 rounds) untested.
7. **Adversarial testing:** Systematically adversarial inputs (jailbreak-style probes) untested.
8. **Non-text modalities:** Text only. Voice/multimodal uncovered.

Total test cases across all experiments: ~942. All data available in source reports. Main paper: CDRA English Paper v2. DOI: [10.5281/zenodo.20968671](https://doi.org/10.5281/zenodo.20968671).